

Beta Decay in Time-Dependent RealQM: A Phase-Coincidence Half-Life and an Energy-Conserving Continuous Spectrum

Claes Johnson¹

¹KTH Royal Institute of Technology, SE-100 44 Stockholm, Sweden ,
claesjohnson@gmail.com

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Abstract

Beta decay is the standard model’s showcase for the weak interaction and, historically, the reason the neutrino was postulated: a two-body decay $n \rightarrow p + e$ would give a monoenergetic electron, yet the observed electron spectrum is continuous, so to save energy conservation Pauli (1930) posited an unseen third particle. We show that in time-dependent RealQM — the complex, current-carrying form of the real-space charge-density formulation, in which the “neutron” is a proton–electron configuration bound by Coulomb alone — all three features of beta decay follow from the dynamics without a weak force and without a neutrino invoked to rescue conservation. (i) The *half-life* arises from *phase coincidence*: each charge domain carries a phase $\theta_k(t) = -E_k t/\hbar$ set by its computed energy, and the loosely bound electron escapes when the inter-domain phase-beats fall simultaneously in an angular escape window; averaged over unknown initial phases this yields exponential survival and a definite half-life fixed by the nucleus’s energy structure. (ii) The *energy question does not arise*, because RealQM is *energy-conserving by construction*: the dynamics is generated by the energy functional as Hamiltonian, so the total energy is conserved at every instant, and there is no deficit for a neutrino to carry. (iii) The *continuous spectrum* is then the natural consequence: the conserved energy is *partitioned* between the escaping electron, the recoiling residual charge, and the radiated field according to the phase configuration at the emission instant, so the electron’s share varies from event to event with a spread up to an endpoint. The original motivation for the neutrino — to save energy conservation — is thus removed, energy conservation holding by construction throughout. We state the scope of the claim, and identify the momentum-recoil balance and direct-detection experiments as the outstanding questions. We are explicit that energy conservation is here a property of the *full* time-dependent equations, while the present numerical results come from a *reduced* phase-coincidence model; a full time-dependent solve of the RealQM equations is identified as the confirming step, and a first radial prototype already confirms the energy conservation numerically (drift linear in the time step, extrapolating to zero).

1 What beta decay asks of a theory

Free-neutron beta decay, $n \rightarrow p + e^- + \bar{\nu}$, confronts any theory with three facts. First, it is *exponential in time* with a definite half-life (mean life ≈ 880 s). Second, the emitted electron has a *continuous* energy spectrum, running from zero up to a sharp endpoint near $Q \approx 0.782$ MeV. Third — and this is the historical crux — a genuine two-body decay $n \rightarrow p + e$ would, by energy

and momentum conservation, make the electron *monoenergetic*; the observed continuum therefore appeared to violate energy conservation. Pauli’s response in 1930 was to postulate a light, neutral, near-invisible third particle carrying off the balance [1], built by Fermi into a quantitative theory of the weak interaction [2].

Two of these three facts are dynamical (the half-life, the spectrum) and one is a conservation argument (the “missing energy”). The standard account addresses all three at once by adding a new force and a new particle. Here we ask what the *time-dependent* form of RealQM says, given that in the RealQM/RealNucleus programme there is neither an elementary neutron nor a weak force to begin with: the model “neutron” is a single proton–electron pair bound by the same Coulomb energy that binds every other nucleus and atom, and this pair is found *unbound*, so its coming-apart into a free proton and a free electron — the charged products of beta decay — is already a Coulomb fact [3]. What the static account leaves open is the *dynamics*: how fast (the half-life), with what energy (the spectrum), and whether energy is conserved along the way. These are exactly the questions the time-dependent form answers.

2 Time-dependent RealQM, and energy conservation by construction

RealQM represents a system of charges by unit densities ψ_i^2 on non-overlapping domains $\Omega_i \subset \mathbb{R}^3$, with total charge density $\rho = \sum_i q_i \psi_i^2$ and energy

$$E[\psi] = \sum_i \frac{1}{2} \int_{\Omega_i} |\nabla \psi_i|^2 + \frac{1}{2} \iint \frac{\rho(x)\rho(y)}{|x-y|} dx dy = \sum_k E_k, \quad (1)$$

which the ground state minimises over the ψ_i and the free boundaries $\partial\Omega_i$; the last equality is the decomposition of the total energy into per-domain contributions E_k . The time-dependent form promotes the ψ_i to complex fields $\psi_i = u_i + iv_i$ evolving by the first-order (Schrödinger) law generated by (1) [5],

$$i \partial_t \psi_i = \frac{\delta E}{\delta \psi_i^*} = -\frac{1}{2} \nabla^2 \psi_i + V_i[\rho] \psi_i, \quad (2)$$

with $V_i[\rho]$ the Coulomb potential felt by domain i from all charges. Equation (2) carries a charge current $\mathbf{J}_i = q_i \text{Im}(\psi_i^* \nabla \psi_i)$ and hence radiation, and it reduces to a stationary state of energy E_k carrying the pure phase rotation

$$\psi_k(t) = \psi_k e^{-iE_k t/\hbar}, \quad \theta_k(t) = -E_k t/\hbar. \quad (3)$$

The point we build on is elementary but decisive. The evolution (2) is generated by the energy functional (1) acting as Hamiltonian, and (1) has no explicit time dependence. Hence *the total energy is a conserved quantity of the motion*,

$$\frac{d}{dt} E[\psi(t)] = 0, \quad (4)$$

exactly, at every instant, for any configuration including one in the act of decaying. RealQM is *energy-conserving by construction*: energy conservation is not an extra postulate to be defended but a structural property of a Hamiltonian dynamics. The same time-translation and space-translation

invariance of (1) makes total momentum (charge plus field) conserved as well.

This already dissolves the historical problem at its root. Pauli introduced the neutrino *in order to save energy conservation* in the face of the continuous spectrum. In a theory where energy is conserved by construction, that motivation cannot arise: there is no instant at which energy goes missing, so there is nothing for an unseen particle to carry off in order to balance the ledger. Whatever energy the initial $p+e$ configuration possesses is present, in full, at every later time — redistributed among the escaping electron, the residual charge, and the radiated field, but never lost. The task is then not to *restore* conservation but simply to read off *how* the conserved energy is shared, and *how fast* the electron leaves.

3 The half-life from phase coincidence

Consider the decaying configuration as a set of charge domains, each rotating at its own phase rate (3) fixed by its energy E_k . Let domain L be the loosely bound electron that will leave. Its escape is not driven by an external force but by a *geometric coincidence* of phases: the electron separates at the first instant when its phase relative to each of the other M domains falls simultaneously within a narrow angular window $[0, \Delta)$ — the window in which the instantaneous Coulomb configuration opens an outward channel across the free boundary. Writing the relative (beat) phases

$$\beta_m(t) = \omega_m t + \varphi_{0,m}, \quad \omega_m = \frac{E_L - E_m}{\hbar}, \quad m = 1, \dots, M, \quad (5)$$

the escape condition is $\beta_m(t) \bmod 2\pi \in [0, \Delta)$ for all m at once. The dynamics is *deterministic*; the only unknown is the set of initial phases $\varphi_{0,m}$, fixed by the configuration's history.

For M incommensurate beat frequencies the fraction of time during which all M phases lie in the window is $(\Delta/2\pi)^M$, and the coincidences occur as a stationary point process in time. The survival probability — the fraction of an ensemble (over initial phases $\varphi_{0,m}$) that has *not* yet met the coincidence — therefore decays at a constant hazard rate,

$$S(t) \simeq e^{-\lambda t}, \quad \lambda \sim \left(\frac{\Delta}{2\pi}\right)^M \langle |\omega_m| \rangle, \quad (6)$$

i.e. *exponential decay with a definite half-life* $t_{1/2} = \ln 2/\lambda$. The rate is a pure function of the nucleus's own energy structure: the beat frequencies ω_m are differences of the per-domain energies E_k that the RealQM solver computes for the actual configuration, and the escape window Δ is set by its geometry. A tightly bound electron (large $|E_L|$, fast beats against many domains) meets the joint coincidence rarely and decays slowly; a loosely bound one decays fast — the qualitative rule that governs beta-decay lifetimes, here with no weak coupling constant anywhere.

A *reduced* numerical realisation of this mechanism — a phase-coincidence model that assigns each domain the rigid rotation (3) and triggers escape on the joint window, rather than integrating the full equations (2) — produces an exponential survival curve $S(t)$ (with good fit) and a corresponding half-life [6]. The essential content is that *randomness enters only through the unknown initial phases*: the exponential law, usually taken as evidence of an irreducibly probabilistic decay, emerges from a deterministic phase dynamics as the ensemble statistics of a coincidence.

That the phase gating is not an artefact of the imposed window can be shown in a genuine solve. A caged electron — a metastable state held behind a barrier, able to leave only by tunnelling — was prepared in a superposition of two quasi-bound levels E_1, E_2 of the well and evolved by a unitary

(Crank–Nicolson) integration of (2) with an absorbing outer layer, *no* escape rule imposed. For a single level the transmitted escape flux is smooth (ordinary Gamow tunnelling); for the two-level superposition it *oscillates* — the electron floats out in bursts — at the beat frequency $(E_2 - E_1)/\hbar$ (reproduced to within about five percent), with several times the modulation of the single-level case. The phase-coincidence gating is thus a real feature of the time-dependent equation, not a modelling device: the relative phase of the caged components opens and closes the escape channel. This is why a *caged* electron does not simply disperse like a free one — it is released, in gated bursts, by phase.

4 The continuous spectrum as partition of a conserved energy

Now the spectrum. At the coincidence instant the electron crosses its free boundary and departs with the kinetic energy it carries at that instant. By (4) the total energy is fixed at the initial value E_0 throughout, so the departing electron’s energy is not free: it is the electron’s *share* of E_0 , the remainder staying with the recoiling residual charge and the radiated field,

$$E_0 = \underbrace{E_e}_{\text{electron}} + \underbrace{E_{\text{rec}}}_{\text{residual nucleus}} + \underbrace{E_{\text{field}}}_{\text{radiation}}, \quad (7)$$

an identity for the full dynamics (2), not an approximation. What varies from event to event is the *partition*. The share E_e is determined by the phase configuration $\{\beta_m\}$ at the moment the coincidence is met: different histories reach the escape window with different instantaneous field and momentum configurations, so the electron leaves with different energies. The result is a *continuous distribution* of electron energies, from near zero (when almost all of E_0 stays behind as recoil and radiation) up to an *endpoint* E_e^{max} (the largest share the geometry allows). The reduced model exhibits a non-zero spread in the released-electron energy — a continuous spectrum, generated by the phase at emission [6] — but, being a surrogate, it *imposes* the spread through a phase modulation rather than *deriving* it from the conserved partition (7). Establishing (7) quantitatively — that an energy-conserving solution of (2) produces this spectrum — requires the full solve discussed in §5.

It should be stressed how *large* the non-electron share must be. In free-neutron decay the electron’s mean kinetic energy is only about 0.30 MeV of the $Q \approx 0.78$ MeV released, the daughter recoil is negligible (below a keV, the proton being heavy), and in the standard account the remaining ≈ 0.48 MeV — the *majority* of the budget — is carried by the antineutrino. For RealQM the same figure sets the bill: the energy that must reside, on average, in $E_{\text{rec}} + E_{\text{field}}$ of (7) is not a small correction but the larger part of Q , and it must fluctuate from event to event to shape the spectrum. This is a demanding quantitative target. A purely two-body ($e + p$) release cannot meet it, since the heavy proton takes essentially none of the energy and the electron would then carry nearly all of Q ; the shortfall between that and the observed mean is precisely the share the radiated field must absorb — which is why the field contribution (§5) is central and not incidental.

The contrast with the standard argument is precise, and it is worth stating carefully because it is the heart of the matter.

- The standard two-body argument fixes the total release $Q = (m_n - m_p - m_e)c^2$ from *particle masses* via $E = mc^2$, treats the products as two point bodies, and concludes the electron energy is *fixed*; the observed spread then signals a missing carrier.

- RealQM makes *neither* assumption. It has no rest-mass energy budget — the release is a *configurational* energy difference read from the functional (1), not a mass difference — and its products are *extended* charge plus field, among which a conserved E_0 is shared. A spread in E_e is then expected, not anomalous, and it coexists with *exact* energy conservation (4).

In short: the continuous spectrum is what an energy-conserving, extended-charge decay *looks like*. Its continuity was read as evidence for a neutrino only under the twin assumptions of a fixed mass- Q and point products, both of which RealQM discards. The endpoint $E_e^{\max}$ plays the role of the standard Q -value, now as the maximal configurational share rather than a mass difference.

5 Scope and open questions

We propose, within time-dependent RealQM and without a weak force, an account of the two dynamical facts (exponential half-life, continuous spectrum) and we remove the energy-conservation motivation for the neutrino (energy being conserved by construction). We are equally clear about what is *not* settled here.

What is solved, and what is not. The energy conservation of §2 is a property of the *full* equations (2): generated by the energy functional as Hamiltonian, they conserve E exactly. The dynamical results of §§3–4, however, are obtained *not* by integrating (2) but from a reduced phase-coincidence model that assigns each domain the stationary rotation (3), and imposes the escape rule and the energy spread by hand; it does not itself enforce conservation. It should be read as an *illustration of the mechanism* — evidence that phase coincidence yields an exponential half-life and a continuous spread — not as a solution of the theory. In the present model, moreover, the per-domain energies E_k of (5) are computed analytically from the configuration geometry rather than read from a per-domain reduction inside the free-boundary solver, and the escape window Δ is a parameter. The decisive next step, now under way, is a genuine time-dependent solve: integrate (2) for the separating configuration, with moving free boundaries, and verify that an actual energy-conserving solution reproduces both the exponential $S(t)$ and the spectrum as the conserved partition (7) — with E_k and Δ then outputs of the solver rather than inputs.

A first step in this direction is already in hand. A radial (spherically symmetric) prototype of the caged proton–electron “neutron” — the electron as a core density $r < a$, the single proton as the surrounding shell $a < r < R$ — evolved under (2) by a staggered real/imaginary leapfrog, confirms the energy conservation (4) numerically: the total-energy drift scales *linearly* with the time step and extrapolates to zero, and it persists under strong perturbation, so the evolution conserves energy up to a controllable discretization error. This prototype establishes only the conservation leg: a fixed radial partition does not let the electron separate, and the directional emission that carries momentum and shapes the spectrum requires the two-dimensional solve. That is the work now in progress.

Momentum recoil — and where exactly it is open. The sharpest empirical support for a third body is not energetic but kinematic: coincidence measurements of the electron and the recoiling daughter show they are *not* emitted back-to-back, so momentum is not balanced by the two visible products alone. A two-dimensional solve sharpens precisely where this bears on RealQM. Evolving the electron and proton as complex fields in a plane, coupled by Coulomb and released from rest by an internal momentum-conserving impulse, the time-dependent dynamics

conserves total momentum to machine precision and produces genuine directional emission with proton recoil: the electron floats off along the axis while the proton, 1836 times heavier, recoils 1836 times more slowly, and the two emerge *exactly* back-to-back ($\mathbf{p}_e = -\mathbf{p}_p$). This is instructive in both directions. It confirms that momentum is conserved by construction and that recoil is real; but it also shows that a clean *two-body* solve *must* give back-to-back products, by conservation alone — so the observed non-back-to-back recoil cannot be reproduced by the two charges by themselves. In RealQM the missing momentum carrier is the *radiated electromagnetic field*: the instantaneous-Coulomb dynamics used here has no field momentum in its ledger, and only the current-carrying (retarded) form of (2) supplies it. The momentum leg is therefore open in a now-precise sense: it lives entirely in the field-momentum contribution, and the decisive test is whether electron, proton recoil, and radiated-field momentum together reproduce the measured recoil–electron angular correlation. It is the point on which the picture will stand or fall.

Direct detection. Inverse beta processes are interpreted as direct neutrino detection (Cowan–Reines). A charge-density account owes an account of these cross-sections; that lies beyond the present, single-nucleus dynamical model.

6 Conclusion

Time-dependent RealQM treats beta decay as the deterministic separation of a Coulomb-bound proton–electron configuration, driven by phase coincidence. The half-life emerges as the ensemble statistics of a joint phase coincidence among domains rotating at their computed energies; the continuous electron spectrum emerges as the phase-dependent partition of a *conserved* total energy among electron, recoil, and field. Crucially, the historical reason for the neutrino — rescuing energy conservation from an apparent deficit — does not arise, because RealQM conserves energy by construction: there is never a deficit to rescue. What replaces the neutrino hypothesis is not a denial of conservation but its opposite — conservation made structural — together with the recognition that the missing “carrier” of the standard picture is the extended residual charge and the radiated field, which a point-body kinematics omits. The outstanding challenge is quantitative: to show that this same field and recoil balance the *momentum* coincidence data as cleanly as the energy ledger balances by construction. That is where the account must next be tested.

References

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